Normalization Proof

BCNF Definition: Arelation RisinBoyce-Codd Normal Form (BCNF) if and only if for every non-trivial functional dependency X → Y, X is a superkey of R.

(1) Category(CategoryID, CategoryName,ParentCatID)
FD: {CategoryID} →{CategoryName, ParentCatID}
CategoryID is the primary key and superkey of
this relation. It is the only determinant, and it uniquely identifies all other attributes. No other non-trivial FD exists. Hence, BCNF is satisfied.

(2) Product(ProductID, ProductName, MRP , Description, Brand, Weight, SKU)
FD: {ProductID} → {ProductName, MRP , Description, Brand, Weight, SKU}
ProductID is the primary key and superkey. It is the sole determinant of all attributes. No partial or transitive dependencies exist. Hence, BCNF is satisfied.

(3) ProductVariant(VariantID, ProductID, Color, Size, VariantSKU, PriceOffset)
FD: {VariantID} → {Color, Size, VariantSKU, PriceOffset, ProductID} VariantID is the primary key and superkey. It uniquely determines all attributes in the relation. No other non-trivial FD exists. Hence, BCNF is satisfied.

(4) Vendor(VendorID, CompanyName, GSTNumber, Email, Vendor Rating, Phone,ContactPerson)
FD: {VendorID} → {CompanyName, GSTNumber, Vendor Email, Vendor Rating, Vendor Phone, ContactPerson}
VendorID is the primary key and superkey. All attributes are directly dependent on VendorID alone. No other non-trivial FD exists. Hence, BCNF is satisfied.

(5) VendorProduct(VP_ID, VendorID, VariantID, SellingPrice, IsAvailable)
FD:{VP_ID} → {VendorID, VariantID, SellingPrice, IsAvailable}
VP_ID is the primary key and superkey. It is the sole determinant of all attributes. VendorID alone cannot determine SellingPrice because the same vendor sells different variants at different prices. VariantID alone cannot determine SellingPrice because the same variant is sold by different vendors at different prices. No partial or transitive dependencies exist. Hence, BCNF is satisfied.

(6) Warehouse(WarehouseID, WarehouseName, City, State, PINCode,Capacity)
FD: {WarehouseID} → {WarehouseName, City, State, PINCode,Capacity}
WarehouseID is the primary key and superkey. All attributes including City, State and PINCode are independently determined by WarehouseID. PINCode, City and State are not dependent on each other as they are separate properties of the warehouse location. No other non-trivial FD exists. Hence, BCNF is satisfied.

(7)Inventory(InventoryID, VP_ID, WarehouseID, Quantity, OrderLevel, BinLocation, LastUpdated)
FD: {InventoryID} → {VP_ID, WarehouseID, Quantity, OrderLevel, BinLocation, LastUpdated}
InventoryID is the primary key and superkey. All attributes are directly determined by InventoryID alone. No partial or transitive dependencies exist. Hence, BCNF is satisfied.

(8) Customer(CustomerID, Email, Customer Name, Phone, CreatedAt) FD: {CustomerID} → {Email, Customer Name, Phone, CreatedAt,PasswordHash}
CustomerID is the primary key and superkey. All attributes are directly dependent on CustomerID alone. No other non-trivial FD exists. Hence, BCNF is satisfied.

(9) Review(ReviewID, CustomerID, VP_ID, Rating, Comment, ReviewDate)
FD: {ReviewID} → {CustomerID, VP_ID, Rating, Comment, ReviewDate}
ReviewID is the primary key and superkey. CustomerID and VP_ID are foreign keys stored as attributes — they do not determine any other attribute within this table. No transitive or partial dependency exists. Hence, BCNF is satisfied.

(10) Address(AddressID, Street, City, State, PINCode)
FD: {AddressID} →{Street, City, State, PINCode}
AddressID is the primary key and superkey. Street, City, State and PINCode are all independent properties of the address and are directly determined by AddressID alone. No other non- trivial FD exists. Hence, BCNF is satisfied.

(11) CustomerAddress(CustomerID, AddressID, IsDefault)
FD: {CustomerID,AddressID} → {IsDefault}
The composite key (CustomerID, AddressID) is the primary key and superkey. IsDefault is a preference specific to a particular customer for a particular address neither CustomerID alone nor AddressID alone can determine it. The only determinant is the composite key which is a superkey. Hence, BCNF is satisfied.

(12) Shipment(ShipmentID, ETA, Status, DispatchDate, Tracking, CourierPartner, InventoryID, OrderID)
FD: {ShipmentID} → {ETA, Status, DispatchDate, Tracking, CourierPartner, InventoryID, OrderID}
ShipmentID is the primary key and superkey. All attributes are directly determined by ShipmentID. InventoryID is stored to track which inventory batch was dispatched — it is a direct property of the shipment, not a transitive one. WarehouseID is not stored directly as it can be derived via the chain Shipment → InventoryID → Inventory → WarehouseID, avoiding redundancy. No other non-trivial FD exists. Hence, BCNF is satisfied.

(13) Cart(CartID, CustomerID)
FD: {CartID} → {CustomerID}
CartID is the primary key and superkey. There is only one non-key attribute which is CustomerID, and it is directly determined by CartID. No other non-trivial FD

exists. Hence, BCNF is satisfied.

(14) CartItem(CartID, VP_ID, Quantity, AddedDate)
FD: {CartID, VP_ID} →{Quantity, AddedDate}
The composite key (CartID, VP_ID) is the primary key and superkey. The same vendor product cannot appear twice in the same cart, so the combination uniquely identifies each cart item. Neither CartID alone nor VP_ID alone can determine Quantity or AddedDate. No other non- trivial FD exists. Hence, BCNF is satisfied.

(15) Order(OrderID, OrderDate, TotalAmount, Status, CustomerID, CouponID,AddressID)
FD: {OrderID} → {OrderDate, TotalAmount, Status, CustomerID,CouponID, AddressID}
OrderID is the primary key and superkey. CustomerID and AddressID are both directly determined by OrderID independently. A customer can place an order to any address including a gift delivery address that belongs to someone else, so AddressID does not determine CustomerID within this table. No transitive dependency exists. Hence, BCNF is satisfied.

(16) WishList(WishListID, CustomerID, CreatedDate, Name)
FD: {WishListID} → {CustomerID, CreatedDate, Name}
WishListID is the primary key and superkey. All attributes are directly dependent on WishListID alone. No other non-trivial FD exists. Hence, BCNF is satisfied.

(17) WishListItem(WLItemID, WishListID, VP_ID, AddedDate)
FD: {WLItemID} → {WishListID, VP_ID, AddedDate}
WLItemID is the primary key and superkey. It uniquely identifies each wishlist item. WishListID and VP_ID are foreign keys — neither alone nor together determine AddedDate independently of WLItemID. No partial or transitive dependencies exist. Hence, BCNF is satisfied.

(18) Payment(PaymentID, OrderID, Payment Status, PaidAt, Amount, TxnRef, GatewayName)
FD: {PaymentID} → {OrderID, Payment Status, PaidAt, Amount, TxnRef, GatewayName}
PaymentID is the primary key and superkey. All attributes are directly determined by PaymentID. TxnRef is a reference number provided by the payment gateway and is a direct property of the payment record, not a determinant of other attributes. No other non-trivial FD exists. Hence, BCNF is satisfied.

(19) OrderItem(OrderItemID, OrderID, VP_ID, Quantity, UnitPrice)
FD:{OrderItemID} → {OrderID, VP_ID, Quantity, UnitPrice}
OrderItemID is the primary key and superkey. It uniquely identifies each line item in an order. OrderID, VP_ID, Quantity and UnitPrice are all directly determined by OrderItemID alone. No partial or transitive dependencies exist. Hence, BCNF is satisfied.

(20) ReturnRefund(ReturnID, Reason, Refund Status, RefundAmount, ReqDate, OrderItemID)
FD: {ReturnID} → {Reason, Refund Status, RefundAmount, ReqDate, OrderItemID}
ReturnID is the primary key and superkey. OrderItemID is a foreign key that directly references which order item is being returned — it is a direct property of the return record, not a determinant of other attributes. No transitive dependency exists. Hence, BCNF is satisfied.

(21) Coupon(CouponID, Code, DiscountType, UsageLimit, ExpiryDate, DiscountValue)
FD: {CouponID} → {Code, DiscountType, UsageLimit, ExpiryDate, DiscountValue}
CouponID is the primary key and superkey. Code is also a candidate key as every coupon has a unique code, making it an alternate superkey. Since both CouponID and Code are superkeys, no BCNF violation exists. All attributes are directly determined by the primary key. Hence, BCNF is satisfied.